Date: 28/09/2026

# Exp-08: Nmap live host discovery ARP, ICMP, and TCP/UDP ping scans.

**Aim:** To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP scan, and TCP/UDP ping scan - and to understand which protocol operates at which layer of the TCP/IP model, and when each one applies.

**Objective:**

- Understand why host discovery must precede port scanning
- Learn how ARP scans work at the link layer using broadcast requests.
- Understand how ICMP echo, timestamp, and address mask requests reveal live hosts across subnets.
- Learn how TCP SYN and ACK ping scans work, and the difference between privileged and unprivileged scanning behaviours.
- Understand how UDP ping scans use ICMP port-unreachable responses to confirm a host is online.

**TryHackMe setup:**

Follow these steps to prepare the lab environment before running any scan:

1. Open the room: https://tryhackme.com/room/nmap01 & log in to your TryHackMe account.
2. Use the embedded network simulator to answer the conceptual questions about ARP broadcast domains.
3. Click Start to deploy the target network - a small lab with one Windows host and one Ubuntu host. Allow up to two minutes for the machines to initialise.
4. Click Start AttackBox to launch the browser-based Kali attack machine in split view next to the room.
5. Once the AttackBox has booted, open a terminal inside it. If instructed by the room, download and apply the provided OpenVPN configuration file so the AttackBox can reach the private lab subnet.

6. Note the Connection - IP / target subnet shown on the room's network diagram — you will substitute this into every nmap command below.

Now your environment is ready to run each host-discovery technique against the lab subnet.

ARP host discovery:

To run only an ARP-based host-discovery scan

```
$ sudo nmap -PR -sn 10.200.6.0/24
```

Sample output:

```
Starting Nmap 7.80 ( https://nmap.org )
Nmap scan report for 10.200.6.1
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in
           10.51 seconds.
```

Icmp host discovery:

Icmp Echo (-PE):

```
$ sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency).
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP addresses (3 hosts up)
           scanned in 3.38 seconds.
```

Icmp Timestamp (-PP):

```
$ nmap -PP -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
host is up (0.069s latency).
Nmap scan report for 10.200.6.250
host is up (0.069s latency).
Nmap done: 256 IP addresses (2 hosts up)
           scanned in 8.44 seconds.
```

Icmp address mask (-PM):

$ nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.14 seconds.

Tcp and udp ping Discovery:

Tcp SYN Ping (-PS):

$ nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.068s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 33.38 seconds.

Tcp ACK Ping (-PA):

$ sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 36.64 seconds.

UDP Ping (-PU):

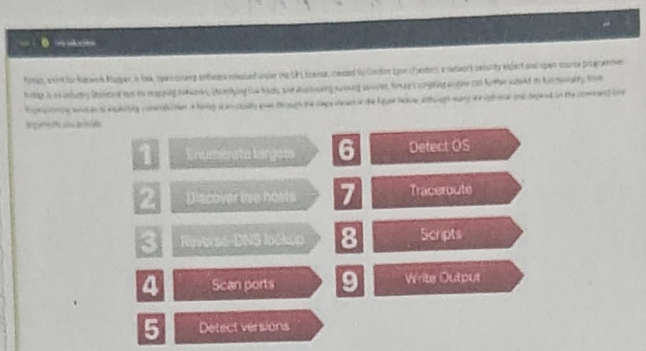
$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 35.97 seconds.

Masscan: Similar discovery concept but transmits packet ofas more aggressively, trading some accuracy ofor raw speed on large ranges.
Its syntax mirrors Nmap's port option.

```
$ masscan 10.200.6.0/24 -p443
$ masscan 10.200.6.0/24 -p80,443
$ masscan 10.200.6.0/24 -p22,25
```

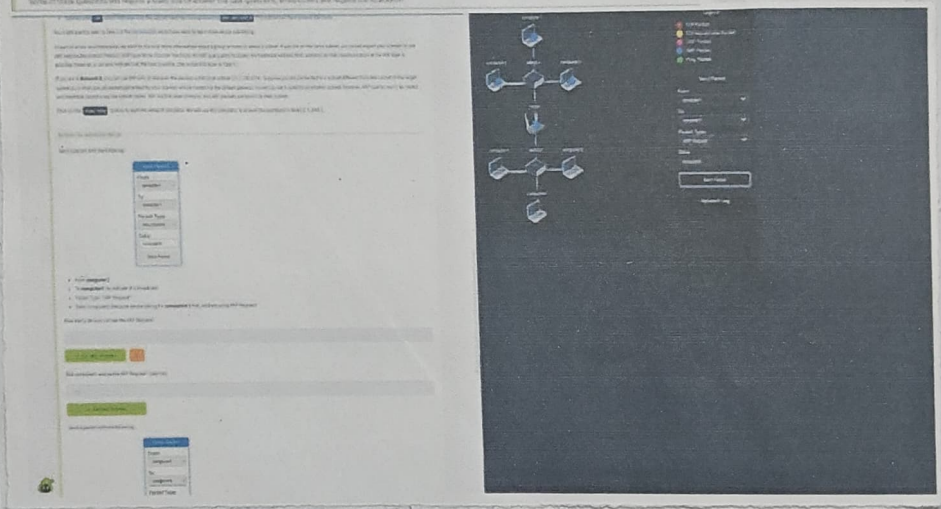
## Command Reference :

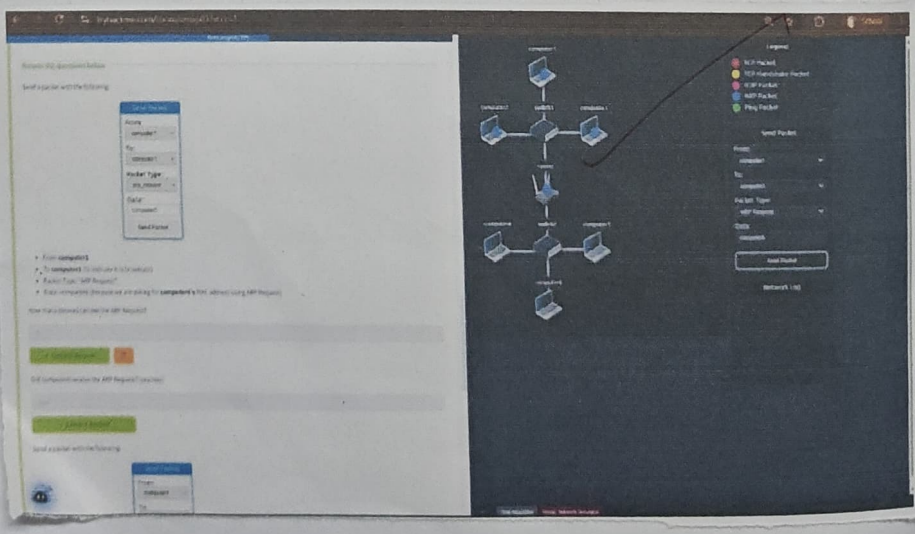
| Option | What it does | Layer / Protocol |
|---|---|---|
| -sn | Host - discovery only; skip port scanning | All (modifier flag) |
| -PR | ARP scan (default & fastest on the local subnet) | Link layer (ARP) |
| -PE | ICMP Echo request / reply | Network layer (ICMP Type 8 / 0) |
| -PP | ICMP Timestamp request / reply | Network layer (ICMP Type 13 / 14) |
| -PM | ICMP Address-Mask request / reply | Network layer (ICMP Type 17 / 18) |
| -PS<ports> | TCP SYN ping (default port 80) | Transport layer (TCP) |
| -PA<port> | TCP ACK ping (needs root / sudo) | Transport layer (TCP) |
| -PU<port> | UDP ping; relies on ICMP port-unreachable | Transport layer (UDP) |
| -n | Disable reverse-DNS resolution | — |
| -R | Force reverse-DNS lookup on every host | — |
| -iL <file> | Read target list from a file | — |
| -sL | List targets only, do not scan / ping them | — |
| -Pn | Skip host discovery - treat all hosts as up | — |

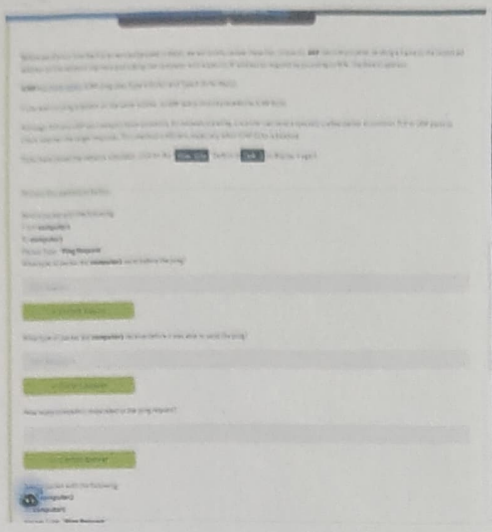


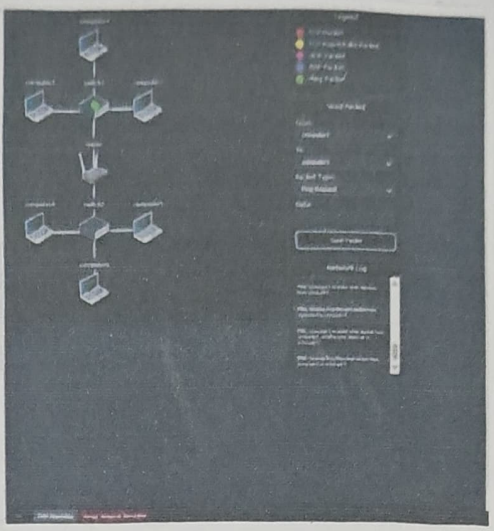

## Learning Objectives

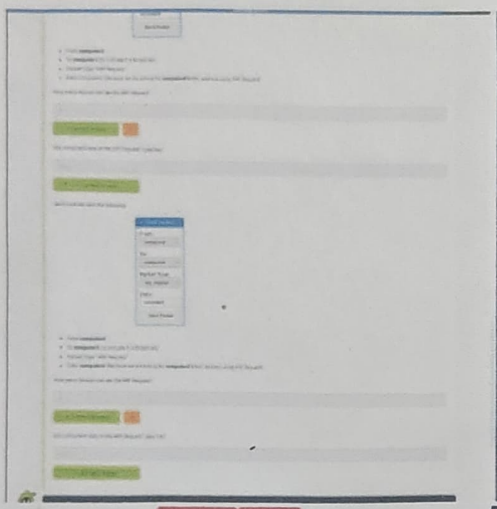
## Learning Prerequisites

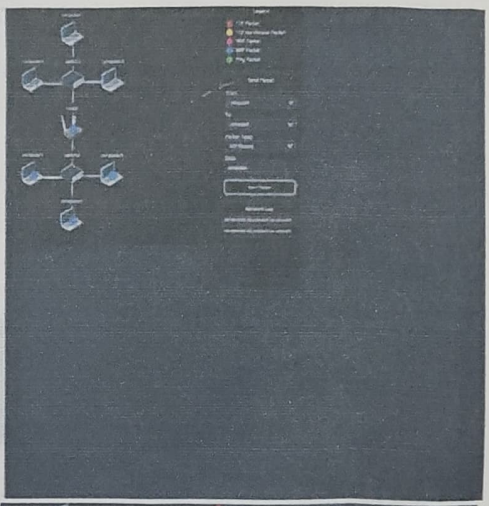


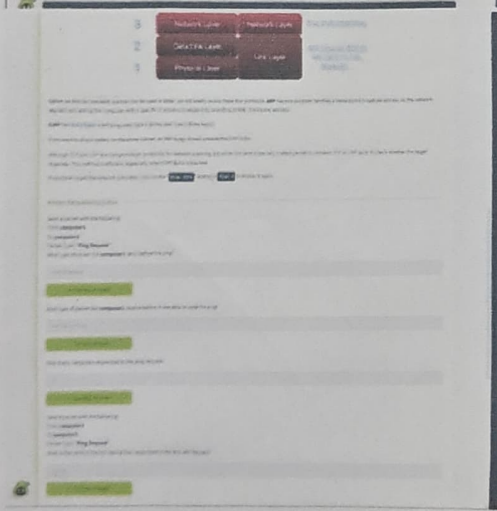


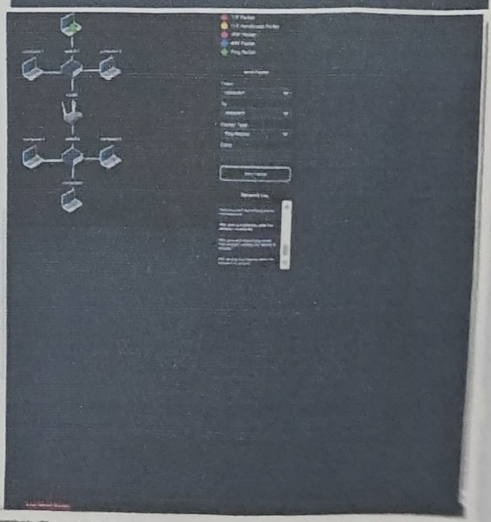












Results: The devices hosts on a network were discovered successfully using Nmap in my hardware environment.